Semantified CEUR-WS in Wikidata

Wolfgang Fahl[1][0000−0002−0821−6995], Tim Holzheim[1,2][0000−0003−2533−6363], Christoph Lange[1,2][0000−0001−9879−3827], Jerven Bolleman[3][0000−0002−7449−1266], and Stefan Decker[1,2][0000−0001−6324−7164]

[1] RWTH Aachen University, Computer Science i5, Aachen, Germany
[2] Fraunhofer FIT, Sankt Augustin, Germany
[3] SIB Swiss Institute of Bioinformatics, Switzerland

Abstract. CEUR Workshop Proceedings (CEUR-WS) has been a non-profit publisher for scientific workshop and conference proceedings since 1995. For more than 30 years it has used HTML and PDF as the single source of truth for the metadata required by indexers such as DBLP and the German national library DNB. Scholars have long craved persistent identifiers for the core entities involved: event series, events, proceedings, papers, editors, authors, institutions, and publishers, with the goal of allowing semantic handling and querying with modern tools.
In this work we show how we enabled the semantification using Wikidata as the target knowledge graph infrastructure. The 2015 Semantic Publishing Challenge queries are finally permanently available with up-to-date metadata. The Scholia portal provides aspect-oriented and user-friendly access and is widely used.
We further point out the challenges of aligning the digital, social, and physical aspects of the ongoing transformation and outline a roadmap to use the available queries to bootstrap a new RDF-based metadata-first publishing approach that will mine metadata from proceedings matter and generate the HTML from it instead of the other way around.

Keywords: Knowledge Graph · Linked Data · Metadata Extraction · Publishing · Semantic Web · Wikidata

1 Introduction

Fig. 1. Most relevant entities for scientific publishing

The Traditional publishing workflows handle the metadata of the needed core entities (as shown in Figure 1) in an unFAIR way. Findable, Accessible, Interoperable, and Reusable (FAIR) 1.2, Single Point of Truth (SPoT), and Single Source of Truth (SSoT) principles are having low priority. Therefore the opportunities to collect the metadata during the different lifcecyle stages of an

event as shown in table 1 are missed. For the future of CEUR-WS publishing we envision a metadata first publishing approach. This work shows how the HTML/PDF SSoT of CEUR-WS at the ceur-ws.org domain (SPoT) has been used to create a Knowledge Graph representation of the relevant entities in Wikidata between 2022 and 2026 as a step in this direction.

Table 1. Scientific Event Lifecycle stages

Stage	**Stakeholders**	**Artifacts Created**	**Format**	**Responsibility**
Announcement	Organizers, Marketing Team	Promotional materials, websites, notifications	HTML, Email	Organizers
Call for Papers (CfP)	Organizers, Authors, Reviewers	CfP documents, emails, submission portals, responses	PDF, HTML, Email	Organizers
Performing Conference	Organizers, Chairs, Presenters, Attendees	Schedules, presentations, participant lists, recordings	PDF, PPT, HTML, Video	Organizers
Proceedings Publication	Editors, Chairs, Authors	Finalized papers, indexed and archived content	PDF, HTML	Editors, Authors, Chairs
Indexing	Libraries, Indexing Services	Catalog entries, metadata records	MARC, XML, RDF	Publishers, Libraries
Referencing/Quoting	Academic Community	Citations, references in future works	BibTeX, HTML	Scholars

Digital Traces of Scientific Events during their lifecyle: Table 1 shows the stages of the scientific event lifecycle that cumulatively create artifacts related to the event.

Moving the responsibility for curating data for relevant entities (see Figure 1) to an earlier stage and thus to the event organizers, while semi-automating the process has several benefits. The quality of the data will increase, and its availability will be much earlier, thus influencing FAIRness—e.g., ensuring computer-readable versions of the core entities are available for downstream systems at each step in the lifecycle. Involving a community such as Wikidata in open source style will further enhance the result [4].

Scientific communication increasingly relies on the use of knowledge graphs, as exemplified by Google Scholar[5] and the evolution of Microsoft Academic Graph [40] into OpenAlex [33]. Legacy systems based on technologies such as MARC [16], PICA [9] or XML are also adapting, highlighted by DBLP's adoption of RDF/SPARQL, with the QLever DBLP SPARQL endpoint[6] [3] being the most up-to-date.

[4] And will create follow-up issues for the single point of truth data to be discussed later

[5] https://scholar.google.com/

[6] https://qlever.cs.uni-freiburg.de/dblp

1.1 CEUR Workshop Proceedings

Since 1995 CEUR Workshop Proceedings (CEUR-WS) (https://ceur-ws.org/) has published more than 4,200 Volumes and over 90.000 papers as a free online service for publishing proceedings of scientific workshops (and smaller conferences) in computer science, operated by at team of unpaid volunteers.

Technically, all data and metadata of the proceedings are directly represented as a static website using HTML, PDF and HTTP and FTP protocols as provided by the workshop editors along the CEUR-WS "how to submit" guide [6].

The metadata is available after some delay of weeks or month through indexing services such as dblp [26] and K10plus[41][7] and at the archive of the TIB `https://www.tib.eu/`.

After multiple unfinished attempts to make the metadata of the CEUR-WS platform available for semantic analysis and querying we report on the successful use of wikidata and scholia as the knowledge graph platform to provide FAIR metadata.

Doing so was harder than one would expect - the devil is in the details.

1.2 Making CEUR-WS FAIR

The FAIR principles [17,42] with Persistent Identifiers (PIDs) and rich metadata as core components are being adopted by an increasing number of publishing organizations.

To make CEUR-WS metadata fair a set of relevant queries should be supported, as derived from the original set of queries of the 2014 Semantic publishing challenge as outlined in Section 2.1. The metadata should be provided following an ontology that is based on established ontologies. The manual and automatic curation of entries should be possible with public access for all stakeholders, e.g., editors, authors, organizers, publishers and indexers. The infrastructure should be stable and there should be sufficient trust in its long term availability An open source non-profit infrastructure is preferred since this is also the mode of operation of CEUR-WS

Given the HTML/PDF/text input of CEUR-WS, the corresponding Knowledge Graph needs to be created and the single-point-of-truth computer readable metadata be separated from the different representations such as HTML so that the above requirements are fulfilled.

Both the HTML and PDF encoding of the original scientific content are structured for the purpose of optimizing the display / output on paper or screens; therefore, there is a structure loss compared to what was originally available in the text document processor files that the authors might have been using [13]. Figure 2 shows the part of the publishing process where the rendering step causes this loss. Most scholars are not aware of this loss in the daily use of published content just because the documents are optimized for display and human consumption [44]. For the metadata extraction and use in knowledge graphs the

[7] `https://dblp.org/`, `https://opac.k10plus.de`

difference is sometimes disastrous, e.g., when a simple text from a PDF document can not be extracted any more due to exotic styling and formatting or simply because only a scanned graphic image version of an older document is available that needs optical character recognition to extract the textual content see table 3.

Fig. 2. Text processing step of digital Scientific Publishing

The metadata needed for creating the knowledge graph is only available in natural language/text form and follows rules that have been changed multiple times during the history of CEUR-WS. From 2013 to 2026, there have been 33 different versions of the index file template with no proper tracing of what was changed from version to version. The index and volume HTML files were often edited manually, leading to minor, undocumented differences even within these versions. The usage of the different page versions follows a long-tail Zipfian distribution with 5 versions covering 60% of all volumes.

1.3 Contributions of this Paper

The core contributions presented in this work are:

1. To provide the tools, infrastructure and approaches to modify the CEUR-WS publication workflow to consistently supply high quality FAIR metadata (Semantification) as detailed in section 5.
2. Splitting the metadata for the three main entities Event, Event Series and Proceedings as a major step towards improving the metadata quality and FAIRness.
3. Providing a bootstrapping [2,7] approach to shift the single point of truth from the current HTML/PDF files to a RDF ready ontology based format.

These contributions are valuable for other publishing use cases see section 6.1.

2 Related Work

2.1 Semantic Publishing challenge

Representing digital traces of scholarly communication in Knowledge Graphs (KGs) [10], [18] is useful for supporting use cases such as literature search and recommendation of events for attendance or publishing. Following the "Semantic Web" [4] approach the Semantification of CEUR-WS has been publicly pursued in the Semantic Publishing challenge [25,34]. Participants were asked to extract an RDF knowledge graph to be able to answer a set of 20 challenge queries.

Kolchin et al. [22,21] have submitted an approach using XPath Queries on the HTML DOM markup and converted the results directly to RDF triples; see ceur-ws-lod repository on GitHub[8]. The reusability of this approach is limited since the parsing and generation code are intermixed.

Sateli and Witte [34] applied the GATE framework [11,12] and a pipeline to create triples from the parsing result [35]; see also their supplementary material[9].

The 2015 work of Milicka and Burget [29] used awk text pattern matching[10] as a tool to parse the table of contents files per volume.

Tasks 2 and 3 called for the detailed analysis of the PDF papers.

All challenge contributions had a purely scientific focus and therefore they would not have been fit for making the results operational and being used in the actual CEUR-WS publishing workflow, as doing so would have made it mandatory to operate and maintain the necessary local self-maintained RDF/SPARQL endpoint to provide a knowledge graph as a point of truth.

For lack of resources (human as well as hardware), and limited trust in the long term stability of such approaches the CEUR-WS team did not make the effort.

2.2 Persistent Identifiers in a scholarly publishing context

A study about Linked Data [19] found that each year about 10% of Linked Data URIs become no longer dereferencable. One way to mitigate the issue is to introduce persistent identifiers (PIDs), which aim to fulfill the following principles [36]: longevity, scalability, extensibility, and security. As noted in [24], the importance to ensure the longevity of PIDs is that "persistence is purely a matter of service". Thus, PIDs can only remain persistent if someone is committed to ensuring that they remain accessible to users. This requires commitment or a service level agreement for PID availability, in contrast to URIs in general, where no such agreement exists.

As [10] notes, PIDs can be resolved via a URI, which follows the first principle of the Den Haag Manifesto from 2011[11]. With this alignment, Semantic Web tools, standards, and concepts are employed to link, map, query, and integrate various data formats and sources. Consequently, knowledge graphs are rendered as usable data adhering to the FAIR data principles..

Franken et al. [15] have promoted the idea of using persistent identifiers (PIDs) for scientific events in the same way as there are already persistent identifiers for papers (DOI), people (ORCID), organisation (GRID, ROR) and books (ISBN). They argue that it is also becoming more and more common practice to use PIDs to identify other important entities or objects. But, as mentioned by Bryl et al. [5], it will only be beneficial if more metadata is provided and the PID is actively used to interlink with other entities.

[8] https://github.com/ailabitmo/ceur-ws-lod
[9] https://www.semanticsoftware.info/sempub-challenge-2015
[10] https://github.com/FitLayout/ToolsEswc/tree/master/awk
[11] `https://doi.org/10.5281/zenodo.55666`

Introducing PIDs in the form of DOIs to CEUR-WS brings up the follow-up problem of who should be responsible for minting the DOIs, when the minting should be done and what the target URL of the DOI should be – not all workshop organizers might like that landing page not to be under their own control. Wikidata's entity identifiers (Q-identifiers) thus promise to be a better alternative, since a rich set of *other* identifiers may be linked to any Wikidata entity, including DOIs, homepages and local and internationally known library and commercial and non-commercial indexing service identifiers.

2.3 Metadata Extraction from PDF, HTML and Text

As part of the Scholia Open Source Project[12], Nielsen [30] created a scraper tool capable of creating QuickStatements [28] output; see scrape/ceurws.py[13]. Using the scrape/QuickStatements chain enables the creation of Wikidata entries for each paper. The Vol-3184/paper4[14] Wikidata entry has been created this way by us to demonstrate the effect.

Proceedings Title Parser[15] [14] has a CEUR-WS parsing mode, which already had the RDFa extractor capability (allowing to cover more recent CEUR-WS volumes using that markup style). Part of this work has been reused and extended to a fully fledged parser in the work we are reporting here.

CERMINE (Content ExtRactor and MINEr) [38] is a software library and a web service[16] for extracting metadata and content from PDF files containing academic publications. The text content is analysed and a structured XML document containing metadata on, e.g., authors and citations is created[17]. The lookup features of CERMINE are limited, e.g., to finding the ISO country code of the country of an institution an author is affiliated with.

GROBID (GeneRation Of BIbliographic Data) [27] has been gradually extended to be a machine learning library for extracting, parsing and re-structuring raw documents such as PDF into structured XML/TEI encoded documents with a particular focus on technical and scientific publications.

We have successfully run CERMINE on 68,524 and GROBID on 69,733 CEUR-WS papers so that the results are now a further potential source for disambiguation according to the original challenge tasks 2 and 3. The corresponding XML files are available via the CEUR-WS single point of truth server see section 5.3.

[12] https://github.com/WDscholia/scholia/issues/1438
[13] https://github.com/WDscholia/scholia/blob/master/scholia/scrape/ceurws.py
[14] https://www.wikidata.org/wiki/Q117040467
[15] http://ptp.bitplan.com
[16] http://cermine.ceon.pl
[17] See CEUR-WS Volume 3352/Paper1.pdf example `https://cr.bitplan.com/index.php/CERMINE/Example`

2.4 Scholarly Metadata in Wikidata

The WikiCite[18] project started in 2014 to work on citation data related to Wikipedia and Wikidata, addressing technical aspects, community building, and the future of open citations. WikiCite is aimed at creating a comprehensive bibliographic database in Wikidata to enhance all Wikimedia projects, with significant progress and community initiatives. As of 2023 the scholarly data in wikipedia is the largest subgraph see Wikidata Statistics[19][1] with 22.6 million scholarly articles providing 31.5% of the 71.6 million instances as of 2023-12.

The Scholia[20] project [30] utilizes the scholarly data in Wikidata to provide detailed profiles for researchers, organizations, journals, publishers, individual scholarly works, and research topics. The Scholia's web portal creates on-the-fly scholarly profiles by querying the SPARQL-based Wikidata Query Service and visualizing the data in various formats. This includes lists of publications, author co-occurrence graphs, topic overlays, and more, significantly enhancing the discoverability and analysis of scholarly communication. Scholia's web portal is an open source project hosted on github[21].

3 CEUR-WS Semantification

3.1 Overview

Figure 3 gives an overview of the CEUR-WS Semantification workflow. The workflow is cyclic – it starts with what has so far been the single point of truth metadata, i.e., the ones embedded in the HTML/PDF/text of the static publication infrastructure. The Metadata Extraction step parses the input files and creates Metadata Records (which are cached as JSON records and in an SQLite relational database). These form the basis for the Matching/Reconciliation that queries Wikidata, DBLP and GND with the respective SPARQL endpoints. The semantified Metadata Records are now available and may be stored in the format we see fit. JSON and YAML are candidate formats see table 3. In the next cycle the parsing of existing records is not necessary any more as long as there have been no changes. When new volumes are published, the main page, listing all proceedings volumes, is modified and HTML tables of content and PDF files are added per volume as submitted by the workshop editors. Currently this is done manually via the tradional workflow and semi-automatic via the single point of truth server see section 5.3.

The steps of the semantification workflow for CEUR-WS as depicted in Figure 3 are explained in the following subsections.

[18] https://www.wikidata.org/wiki/Wikidata:WikiCite
[19] https://www.wikidata.org/wiki/Wikidata:Statistics
[20] https://scholia.toolforge.org
[21] https://github.com/WDscholia/scholia

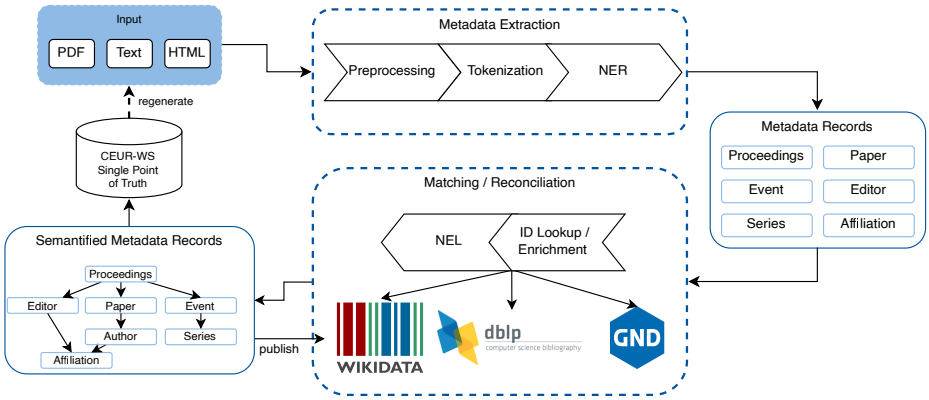

Fig. 3. Workflow of the CEUR-WS Semantification

3.2 Preprocessing

To extract the relevant markup elements, we created parsers for the index and volume HTML Files as part of the pyCEURMake project. These parsers handle the RDFa-like annotations that have been applied in newer CEUR-WS volumes as well as the special and exotic cases arising from the long tail follow-up problems of the different over 33 styles of HTML structure being used. The BeautifulSoup4 Python library is used for lenient HTML parsing as a basis.

3.3 Tokenization

Disambiguation using Event Signatures As outlined in Section 2.2, PIDs would be useful for uniquely identifying scientific events. In the absence of PIDs, it is necessary to use a quasi-identifier [31] consisting of a set of metadata elements referred to as "Event Signature". There is neither a standardized definition of event signatures nor a recommendation for their use in references and proceedings titles. Retrieving the signature from the volume's textual description is a core step in creating the CEUR-WS knowledge graph.

As part of [15], the first author of this article has shown that a typical scientific event "signature" consists of the following metadata (the example event being ISWC 2019[22] *The Semantic Web – ISWC 2019: 18th International Semantic Web Conference, Auckland, New Zealand, October 26–30, 2019*):

acronym a short name for the conference, often consisting of 3 to 8 upper case letters aiming at uniqueness but actually still often being ambiguous. For instance, ISWC may refer to the **International Semantic Web Conference** or to the International Symposium on Wearable Computing.

frequency annual, biennial, triennial – most events have an **annual** frequency and this is mostly **not stated explicitly**.

[22] https://www.wikidata.org/wiki/Q48027931

event reach target reach of the conference such as **international**, European, East Asian

event type such as **Conference**, Workshop, Symposium

year a two or four digit reference to the year in which the event took place – not to be confused with the year of publication of the proceedings, which might be different (**2019**)

ordinal often used to enumerate the conference series instances (**18th**)

date start date and end date or date range of the conference (**October 26–30**)

location description of the location of the conference often consisting of country, region and city – sometimes with details about the exact venue. (**Auckland, New Zealand**)

title the title often contains scope, type and subject of the conference (**International Semantic Web Conference**)

subject description what the conference is about, often prefixed with "on" (**Semantic Web**)

delimiters a variety of syntactic delimiters such as blanks, comma, colon, brackets are used depending on the citation style.

The event signature needs to be extracted from the CEUR-WS main and volume tables of contents and stored as triples for the target KG.

The distinction between proceedings, event and event series needs to be made – therefore the result needs to be split and disambiguated against existing entries in the target KG.

The mapping as outlined in Table 2 has been used to map to the "event" and "proceedings" entry in Wikidata. The top rows of the table show the common properties of the proceedings and event item entries, followed by the special properties for events followed by the special properties for proceedings.

The series entry for the Text2KG example has been created and "colocated" property that has been added as part of this work recently is filled for this example. The Text2KG Workshop series Scholia overview[23] shows the connections.

As an example, we are using the Wikidata entry for the Text2KG@ESWC-2022[24] workshop, whose proceedings have been published as CEUR-WS Volume 3184[25] with the Wikidata proceedings item being shared by another workshop (a special but still frequent case required by the CEUR-WS submission guidelines).

3.4 Named Entity Recognition and Linking (NER & NEL)

The Named Entity Recognition (NER) and Named Entity Linking (NEL) tasks for the CEUR-WS Semantification are based on the textual input from the HTML markup that needs parsing into tokens that represent entities, and then matching the textual content of the tokens against Wikidata entries that might need disambiguation.

[23] https://scholia.toolforge.org/event-series/Q116982161
[24] https://www.wikidata.org/wiki/Q113512465
[25] https://ceur-ws.org/Vol-3184/

Table 2. Mapping Event Signature Elements to Wikidata

property	PID	example
acronym	short name (P1813)	Text2KG 2022
title	title (P1476)	1st International Workshop on Knowledge Graph Generation From Text
event type	instance of (P13)	Workshop
date(start)	start time (P580)	May 30th, 2022 → 2022-05-30
date(end)	end time (P582)	May 30th, 2022 → 2022-05-30
location	location (P276)	Hersonissos → Chersonesos Irakliou (Q1018106)
country	country (P17)	Greece → Greece (Q41)
series	part of the series (P179)	International Workshop on Knowledge Graph Generation From Text → Workshop on Knowledge Graph Generation From Text (Q116982161)
ordinal	series ordinal (P1545)	1st
colocated with	colocated with (P11633)	ESWC 2022 → ESWC 2022 (Q110791806)
homepage	official website (P856)	https://aiisc.ai/text2kg/
URN	URN-NBN (P4109)	urn:nbn:de:0074-3184-1
publication date	publication date (P577)	2022-08-11
DBLP id	DBLP publication ID (P8978)	conf/esws/2022text2kg
k10plus id	K10plus PPN ID (P6721)	1818588285

The semi-structured HTML markup is simpler to handle than natural language processing of plain text since the parsing of the different entity types can be guided by structural hints such as expecting a title in an `h1` HTML tag context and therefore increases the accuracy of the matching process [8]. Items to disambiguate are derived from the event signature as outlined in section 3.3: Volumes, Papers, Editors, Authors, Locations (Country/Region/City), Dates, Ordinals, Acronym, Homepage.

For the phase of the project we are reporting, the mass creation of Proceedings and Event entries was in the focus. The Paper, Editor and Author disambiguation and Event series completion has been prepared and example results are available to show that the elements are available and may be systematically created and queried.

Location NER and NEL The location of an event is described in the table of contents in *span* elements usually classified as *CEURLOCTIME*. Their value contains semi-structured information about the event's location and date range, e.g., "Hersonissos, Greece, May 30th, 2022". Besides the varying formats of the location and date definition, the location information can be fairly easy separated from the date. This leaves a string that should contain information about the city and country. There exist cases where also the region or venue is named, and since more and more conferences have moved to virtual meetings since 2020, the location string could also contain indications for that. Since location strings can be identified on extraction, Named Entity Recognition (NER) and Named Entity Linking (NEL) are done in one step to get Wikidata QIDs of the mentioned locations. For the NEL we used geograpy3, a Python library that has a database with the labels of countries, regions and cities in multiple languages linking to the corresponding Wikidata QID. The response of this label lookup is a list of possible locations of the aforementioned categories. The list is sorted by the category order city, region, country where the cities are also ordered by population. To this list, we once more apply a ranking, since we know that in most cases the country is named within the string, so we can select the city in the given country context. For the given example the result would then be "Hersonissos, Greece" → Chersonesos Irakliou (Q1018106)[26] (Greece (Q41)[27]).

Editors and Authors The author and editor name disambiguation is one of the main challenges for libraries and indexers [37]. Due to the common occurrence of duplicate names, abbreviations of first names, typos and encoding errors disambiguation is an expensive and error-prone task if high accuracy is aimed for.

In CEUR-WS, the editor information is given in an HTML element containing the editor signature, usually given name and family name with a numeric reference to the affiliations, as shown in Listing 1. For newer publications, OR-

[26] https://www.wikidata.org/wiki/Q1018106
[27] https://www.wikidata.org/wiki/Q41

```
<b> TEXT2KG edited by </b>
</p><h3>
   <span class="CEURVOLEDITOR">Sanju Tiwari</span> ... 1
   <span class="CEURVOLEDITOR">Nandana Mihindukulasooriya</span> ... 2
   <span class="CEURVOLEDITOR">Francesco Osborne</span> ... 3 4
   <span class="CEURVOLEDITOR">Dimitris Kontokostas</span> ... 5
   <span class="CEURVOLEDITOR">Jennifer D'Souza</span> ... 6
   <span class="CEURVOLEDITOR">Mayank Kejriwal</span> ... 7
</h3>
```

Listing 1: CEUR-WS volume page HTML markup excerpt of editor definition

CIDs of Authors and Editors might be directly available from the PDF input. For older volumes, only the plain name, affiliation and no identifier is provided that could simplify the disambiguation. Fortunately, 78% of the volumes are indexed at DBLP – less than 800 early volumes are missing. DBLP provides high quality disambiguated data about the proceeding editors and paper authors also accomplished through manual curation [20]. Therefore, extracting the editors and resolving each name to an identifier by looking up the DBLP id seems to be the best option. The same strategy as used for the editors also applies to the authors but here the affiliation needs to be extracted from the paper PDF in the future.

3.5 ID Lookup/Enrichment - DBLP, k10plus Volume matching and linking

DBLP and k10plus records may be trivially matched by the unique identifier volume number of a proceeding combined with the URN of the CEUR-WS proceedings series.

We are using the QLever DBLP SPARQL endpoint mentioned in Section 1 to match CEUR-WS volumes against DBLP entries by volume number.

For the k10plus matching we use the catmandu library, which allows to query the PICA [9] (a library cataloging format used predominantly in German-speaking countries) based k10plus database for URN matches; for details, see the PPN/Volume/WikidataItem matching SPARQL Query[28].

Based on the resolved IDs from the ID lookup and NEL, we can now enrich our data by querying additional or missing data. For example, in Volume 3356[29], only "Tokyo" is defined as location; linking the string to Tokyo (Q1490)[30] then enables querying for the missing country information. Similar enrichments are done for editors and authors to complete the records.

[28] https://w.wiki/6Qm5
[29] https://ceur-ws.org/Vol-3356/
[30] https://www.wikidata.org/wiki/Q1490

3.6 Decision to use Wikidata as a target

Wikidata [39] is a knowledge graph based on an RDF triple store that has been successfully used to gather and link metadata of scholarly communication artifacts [30].

In 2022, we decided to directly target Wikidata instead of trying to set up our own RDF/SPARQL endpoint, as outlined in Section 2.1. Wikidata is well suited to to handle the challenges listed in Section **??** as referenced below by priority:

- **Unique Identifiers (F1)**: by assigning Q-Identifiers, such as Q5 for human.
- **Rich, Identifiable Metadata (F2, F3)**: by being originally based on Wikipedia entries from multiple countries and using a very large community and bots to add current items.
- **Searchable Data (F4)**: by providing a robust Wikidata Query Service .
- **Curation and Accessibility (A1, A2, R1.2)**: by enabling community edits and offering multiple language support.
- **Complex Queries Support (A1, I1, I2, I3)**: by facilitating sophisticated querying through a SPARQL endpoint which may be federated with other Linked Open Data endpoints.
- **Ontology Reuse (I1, I2)**: by integrating and linking to external ontologies and databases and providing its own robust ontology curation process.
- **Stability and Trust (R1.3)**: by ensuring stability and trust through Wikimedia Foundation governance and having mirrored data in for profit (e.g. google) and non profit organizations[31].
- **Open Source, Non-Profit (R1.1)**: by promoting an open-source, non-profit model with Wikimedia Foundation and community contributions.

3.7 Workshop colocated with conference

Most CEUR-WS Volumes are proceedings of workshops, whose majority is colocated with a conference. For this "colocation" relation there was no specific property in Wikidata when we started the semantification. Kolchin [21] had already pointed out that "BIBO doesn't have an 'event is part of bigger event' semantics" in 2015, so the need was long known. We initiated the creation of P11633 (colocated with)[32] by a property proposal[33] according to the Wikidata's property proposal process[34] which states "When after some time there are some supporters, but no or very few opponents, the property is created by a property creator or an administrator."

[35].

[31] e.g. RWTH Aachen i5 runs multiple copies of Wikidata!

[32] https://www.wikidata.org/wiki/Property:colocated with

[33] https://www.wikidata.org/wiki/Wikidata:Property_proposal/colocated_with

[34] https://www.wikidata.org/wiki/Wikidata:Property_proposal

[35] One further criterion was that at least three examples need to be supplied. Unfortunately we did present a few dozen examples but not in the format expected which held up the process by a few months

After that, there was a lively and productive discussion that lead to the clarification that the property is asymmetric. The property was considered as highly relevant and well defined. After almost a year of preparation and discussion the Property is now available and shall be used in the future.

4 SemPubFlow

4.1 Overview

SemPubFlow represents a transformative approach in the lifecycle of scientific events as outlined in Table 1, advocating for early curation and integration of publicly available machine-readable metadata separating the concerns of content and visualization.

Traditionally the created artifacts are mostly intended for human consumption and therefore optimized in this respect. Table 3 shows the computer readability challenges of some of the most common formats which often need sophisticated natural language processing capabilities to be performed. A limiting factor is the lack of standardization and separation of concerns. With the advent of LLMs the options for processing these diverse sources of metadata are vastly improving—making it feasible to get computer readable results as early as possible.

Format	**Description**
Plain text	Requires natural language processing (NLP) for metadata extraction due to its unstructured nature.
HTML	Readability varies with the structural and semantic rigor in the document. Microformats and RDFa are sometimes available.
PDF	Primarily being a collection of individual glyphs to be positioned and formatted, pose challenges in text reconstruction and structure analysis.
BibTeX	Structured nature makes it highly suitable for computer readability and metadata extraction in scholarly communications.
XML/JSON/YAML	Markup languages that balance human and machine readability with varying degrees of structure strictness.
RDF	A standard model for data interchange, facilitating data merging and schema evolution support across diverse schemas.
Multimedia (Videos, Pictures)	Visual media often require sophisticated processing such as OCR for text extraction a for content extraction.
Other (PPT, Word, SVG, etc.)	Vary in readability; structure and semantics can usually be accessed via specific APIs or tools

Table 3. Computer Readability of various formats used for scholarly artifacts

4.2 Metadata first semantic publishing

A key goal of metadata first semantic publishing is the early use (and enforcement of this use) of persistent identifiers. Without such identifiers it is notoriously difficult to disambiguate the core entities involved in scholarly publishing.

The following PIDs are already often in use but the enforcement of their use is lacking in CEUR-WS.

- **ORCID (Open Researcher and Contributor ID):** Identifier for individual researchers.
- **ROR (Research Organization Registry):** Identifier for research organization.
- **Wikidata-ID:** A general identifier for entities in the Wikidata database, often making further identifiers available for reference.
- **DOI (Digital Object Identifier):** A general identifier for objects accessible on the internet – often used for scholarly papers.

We propose to make the use of PIDs mandatory for publishing in CEUR-WS. Given the existing index entries of CEUR-WS Volumes in DBLP and the German national library, the DBLP and GND identifiers are also candidates. For ORCIDs, RORs, DOIs there is active action required by the stakeholders to obtain these, and, in the case of DOIs, there is cost involved. Using Wikidata as the general PID Infrastructure is much more feasible since it offers a freely accessible, community-driven platform that PIDs for a wide range of entities. This approach aligns with the open and collaborative nature of scholarly communications and meets the requirements for easy integration, broad adoption, and maintenance. Using Wikidata Q-Identifiers as PIDs relies on the long term availability of Wikidata which is quite likely giving the support by a strong community and big commercial players such as google.

The Wikidata PID usage proposal calls for looking up or creating Wikidata entries for all relevant core entities as early as possible and linking these entries to the standard PIDs where applicable.

In the workflow the existing entries will later be reusable for computer assisted lookups as part of the SemPubFlow tool see section **??** and Scholia.

Figure 4 shows the proposed future workflow. Publication via CEUR-WS would ideally involve assigning the Volume number and transferring the metadata-rich candidate snapshot to the Single Point of Truth. This process ensures the completeness and quality of the metadata allowing for the generation of traditional HTML pages and the official stamping as a CEUR-WS proceeding.

Fig. 4. Metadata First Scholarly Publishing Workflow Overview

4.3 Iterative introduction

The software engineering challenge in introducing the SemPubFlow workflow is to find a sweet spot between effort and benefit [32]. Given the long-tail distribution of issues of ever more corner and exotic cases that have shown up in the analysis of the existing data and workflow there is potentially a huge list of "small" problems that would require an inadequately high amount of effort to be solved systematically. The "sweet spot" is in solving only the most relevant problems and not fixing some of the exotic details at all or doing it manually. Therefore we will avoid a "big bang" introduction, which would replace the old workflow with the new SemPubFlow in one single big (and therefore very risky) step. Instead we propose to do smaller changes gradually. As a first step, the main index.html shall be split by years (with a link to the old legacy complete list). Then, the per-year list may be partly created using the new SemPubFlow approach. This will be continued iteratively, going from the current year back to a point where it seems not reasonable any more to spent further effort.

4.4 Legal aspects of Publishing Personal Data of Scholars

The proposed workflow calls for publishing personal data of authors and editors early in the life cycle of an event.

This is perfectly legal under at least one of the following conditions:[36]

1. **Consent:** Explicit consent from individuals for their data to be published – e.g., by signing an authors agreement.
2. **Legitimate Interests:** Publication serves legitimate interests that outweighs the individual's rights and freedoms. E.g., when the scholar is a person of public interest.
3. **Public Data:** Data has already been made public by the scholar.
4. **Scientific Research:** Processing is necessary for scientific research purposes as per Article 89 of the GDPR.
5. **Contractual Necessity:** Necessary for the performance of a contract with the scholar – e.g., publishing a paper in workshop proceedings and making the metadata available for indexing by Wikidata, dblp, libraries and other interested parties.
6. **Legal Obligation:** Necessary for compliance with a legal obligation. This would, e.g., be the case if the publication was done via a traditional print outlet and thus, by German law, a copy of the proceedings has to be made available to the German National Library. On indexing, the personal data records of authors will be created as part of the German National Library's legal commitment.
7. **Vital Interests:** Necessary to protect the vital interests of the scholar or another person. This category is more relevant to medical or emergency situations and rather unlikely to be applicable for the scholarly publication use case.

[36] Disclaimer: this is just a summary in layman's terms of legal analysis that was done as part of the CEUR-WS semantification project

8. **Public Interest:** Necessary for performing a task carried out in the public interest or in the exercise of official authority. E.g., when an investigation by public authorities is underway.

Consent, Public Data, and Contractual Necessity are crucial for scholarly publishing. Considering the analogy with Legal Obligations applying to print outlets, we expect it to be feasible to convince scholars that publishing their necessary personal data is reasonable in digital contexts.

5 Implementation and Demos

5.1 Prototypes

A prototype for the presentation[37] of the CEUR-WS Semantification results has been created using Semantic MediaWiki [23] (SMW), which is using the same MediaWiki open source engine as Wikipedia. SMW has extensions for markup that is transforms subject-predicate-object statements to triples, which leads to a "Semantification" of the Wiki by making the triples available for query. Like most SMW installations we are using the standard SQL based triplestore and ask queries and not RDF/SPARQL [43].

A GitHub project for the single-point-of-truth metadata handling and conversion to different representations has been started at ceurws/ceur-spt[38].

Further background research material is supplied via the Semantic MediaWikis for Wolfgang Fahl's PhD[39] (public) and the ConfIDent requirements wiki[40] (access on request).

5.2 CEUR-WS Volume Browser

Figure 5 shows a screenshot of the CEUR-WS Volume Browser, which we created as a means to support semantification tasks such as transferring metadata of recently added volumes to Wikidata as well as showing the available index entries for volumes that have been published for a few weeks/months already. The example shown is for CEUR-WS Volume 3262 Wikidata Workshop 2022[41].

Figure 6 shows an enlarged section of the screenshot where the links between the proceedings volumes and the external knowlede graphs are presented. In the example, these are wikidataitem[42], dblp[43], k10plus[44], and the Scholia links to the proceedings, event and event series[45] (which has links to the event and proceedings).

[37] http://ceur-ws.bitplan.com
[38] https://github.com/ceurws/ceur-spt
[39] https://cr.bitplan.com/index.php/Category:Text2KG
[40] http://rq.bitplan.com
[41] https://ceur-ws.org/Vol-3262/
[42] http://www.wikidata.org/entity/Q115053286
[43] https://dblp.org/db/conf/semweb/wikidata2022
[44] https://opac.k10plus.de/DB=2.299/PPNSET?PPN=1830580760
[45] https://scholia.toolforge.org/event-series/Q106429025

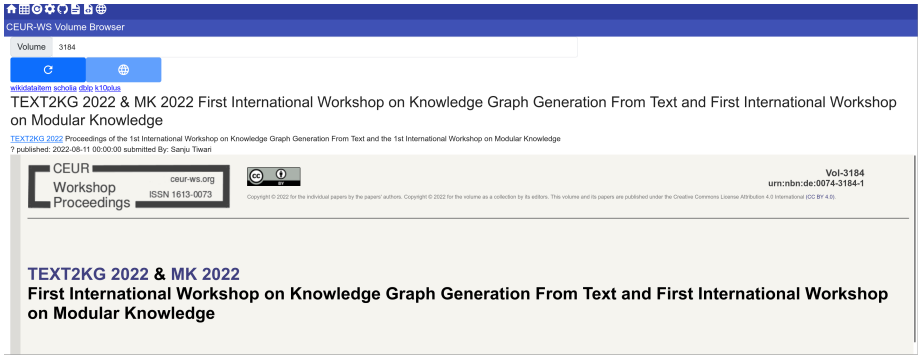

Fig. 5. Screenshot of the CEUR-WS Volume Browser

Fig. 6. CEUR-WS Volume Browser enlarged details with links to external KGs

5.3 CEUR-WS Single Point of Truth Server

The Single Point of Truth Server serves as a prototype and proof of concept for the CEUR-WS semantification. Instead of static HTML, all content displayed by this server is generated from the semantified metadata. The separation of concern of metadata and presentation is implemented here.

The server software is available open source at github[46]. It runs on uvicorn ASGI server and FastAPI web framework and has only a limited list of dependencies to stable Python libraries for long term maintainability.

Key features are HTTP content negotiation for human (HTML) and computer (JSON,YAML,XML, ...) consumption and a documented RESTFul API.

The demo of the server is available at the RWTH Aachen i5 chair.

Using FastAPI version 0.1.0 with Swagger OAS3[47] compatibility, a comprehensive set of endpoints is offered. There are endpoints which are compatible with the traditional static for index and volume HTML tables of contents, as well as paper PDFs that deliver HTML content. In addition various other computer readable representations are available.

Notably, each paper is accessible via a dedicated endpoint, a feature absent in the traditional static CEUR-WS implementation. Links to corresponding volumes and author indices (dblp, GND, and Wikidata) are provided, while granting direct access to Scholia profiles based on the Wikidata items.

[46] https://github.com/ceurws/ceur-spt
[47] https://swagger.io/specification/

The paper endpoint enables retrieval of both the PDF and its plain text. It integrates the CERMINE and GROBID results for advanced XML analysis. The metadata can be exported as QuickStatements for Wikidata, wikibase-cli, Semantic MediaWiki Markup, or in standard formats such as JSON or YAML, facilitating interoperability and analysis for different use cases. Individual entity imports and amendments for different target platforms are thus possible. This is a step towards the scholarly CMS concept presented in section 6.1.

Figure 7 shows an example paper from the second TEXT2KG@ESWC2023 workshop.

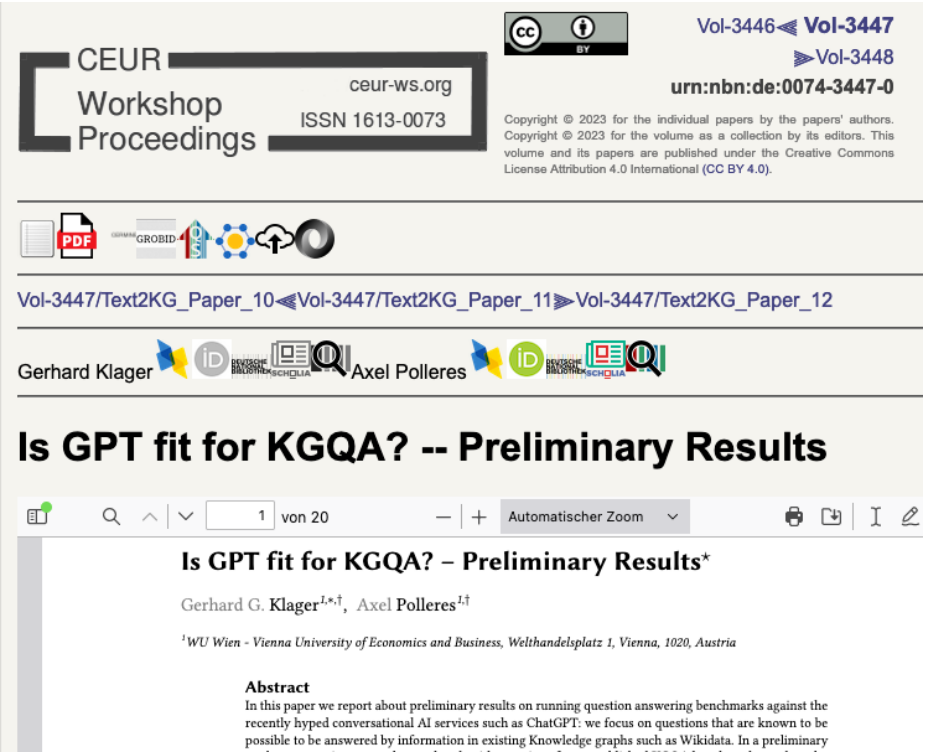

Fig. 7. Paper landing page of CEUR-WS Single Point of Truth server

When scholars change their name over time they are often inclined to ask to have their name changed in the original publications. While this is mostly not feasible the paper endpoint approach with an author bar allows to display the new name.

5.4 Single Point of Truth against Wikidata validation

A major concern of the CEUR-WS editors and other stakeholders is the trustworthiness of the Wikidata entries. These entries are mostly created by the CEUR-WS Wikidata bot but also by other tools such as Scholia and individual Wikidata curators. Therefore, it is necessary to regularly check that the single point of truth data is consistent with the Wikidata entries. This is especially important for the identifiers being used. The pyLodStorage synchronization tool (see section **??**) checks the consistency. Table 5 shows an example result for the URN-NBN persistent identifier. In 580 cases the URN identifier is set in Wikidata (since the URNs can be calculated from the volume number as explained below) but not available from the volume pages via a microformat annotation such as

```
<span class="CEURVOLNR">Vol-3601</span>
```

as was the case for very early volumes.

In 17 cases, there was a mismatch originating from mistakes in the manual execution of the traditional workflow. These cases have been checked manually and could be partly fixed via the wikibase-cli command line tool with generated fixing commands such as:

```
$ wd add-claim Q116525021 P4109 "urn:nbn:de:0074-3249-2"
```

However, some of the URN entries where wrong in the input Volume HTML files see Table 4. Finding these problems was notoriously difficult since the documentation of DNB URN-NBN check digit generation is obscure and the website tool for generating the check digits does not have an API (see CEUR-Make issue 88[48] for the details) [49].

Volume	URN
Vol-2018	urn:nbn:de:0074-2018-6
Vol-2210	urn:nbn:de:0074-2212-3
Vol-2479	urn:nbn:de:0074-2479-C
Vol-2743	urn:nbn:de:0074-2743-1
Vol-2843	urn:nbn:de:074-Vol-2843-4
Vol-3419	urn:nbn:de:0074-3419-9
Vol-3466	urn:nbn:de:0074-XXX-1

Table 4. URN checksum Check

[48] https://github.com/WolfgangFahl/pyCEURmake/issues/88

[49] different versions of the tool in Javascript, PHP and python needed to be compared due to the C-puzzle book style handling of pos++ in the code.

Table 5. URN-NBN Synchronization

left ↔ right	#	%
CEUR-WS ← Wikidata	580	16.10%
CEUR-WS ↔ Wikidata	3006	83.43%
CEUR-WS → Wikidata	17	0.47%

5.5 Metadata Query capability

Having the CEUR-WS metadata available in Wikidata allows for standard SPARQL queries, e.g., using the Wikidata Query Service, to be applied to analyze it. Figure 5 shows a map of the distribution of event locations created with such a query. The relevance of the original set of 20 queries for Task 1 of the Seman-

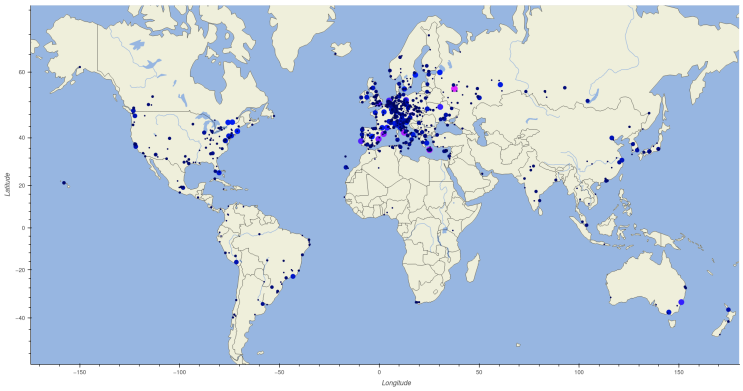

Fig. 8. Locations of all CEUR-WS proceedings events (Query 0.10)

tic Publishing Challenge, which were set as a benchmark in 2014 for different stakeholders, was subjectively rated by us, resulting in a list sorting the queries by priority[50]. The most relevant queries and the 5 queries Q1.5, Q1.12, Q1.13, Q1.16 and Q1.17 that rely on the main index have been implemented as SPARQL queries[51] that are compatible with the Wikidata Query Service endpoint to prove that our approach covers the intentions of the original challenge. Our result supplies even more capabilities given the option to run federated SPARQL queries over the connected Wikidata, DBLP and k10plus knowledge graphs. The use of Wikidata IDs as persistent identifiers is a core success factor here.

[50] https://cr.bitplan.com/index.php/List_of_Queries
[51] https://cr.bitplan.com/index.php/Semantic_Publishing_Challenge_Queries

5.6 Evaluation by Further Quality Metrics

Making the CEUR-WS Volume metadata available on Wikidata has improved the indexing **coverage** to 100% of all valid Volumes compared to 69% for k10plus and 76% for dblp.

The **timeliness** of the CEUR-WS metadata in Wikidata is much higher than for DBLP or k10plus. For DBLP it takes a few days to weeks, for k10plus it may take weeks to months before the metadata shows up. The Wikidata update may be done immediately when publishing with no delay - with SemPubFlow the data will be available as early as the event organizers see fit. With the separation of event and proceedings entries it is now possible to show future events for which there are no proceedings available yet as soon as the events have been announced, and later link the detailed proceedings metadata to the event record.

6 Conclusion

We have presented the first steps of CEUR-WS Semantification that result in the metadata of CEUR-WS Volumes being available in Wikidata. The linking of the relevant entities for workshops, the conferences these workshops might be colocated with, the event series that these workshops and events might form, as well as the linking to editor, author and paper entries and the affiliated institutions has been prototyped.

The cross-linking with DBLP and k10plus has been performed and may now be continuously applied in the future.

Given that all four involved meta data sources – CEUR-WS, Wikidata, DBLP and k10plus – involve a lot of manual curation, data quality errors deriving from human errors still have to be mitigated with the goal to achieve a lower error rate than would be possible with manual efforts alone.

6.1 Future Work

Figure 9 shows an overview of a possible new approach to publishing via CEUR-WS. The core idea is to separate the concerns of displaying content (such as static HTML/Semantic MediaWiki) from the storage of metadata in a knowledge graph, e.g., Wikidata.

The new publishing workflow shall be based on single-point-of-truth metadata that is kept in a computer readable format such as JSON and generate the HTML presentation from this metadata.

CEUR-WS papers do not have DOIs assigned to them during the publishing process. Assigning DOIs to papers is a feature much asked for by workshop organizers these days, which CEUR-WS did not supply in the past. The new approach/architecture would simplify the DOI minting, since the necessary metadata is a subset of the metadata we intend to provide anyway.

The first phase shows the current state of the workflow in the prototype phase, which we report on in this paper, while the second phase is the goal of

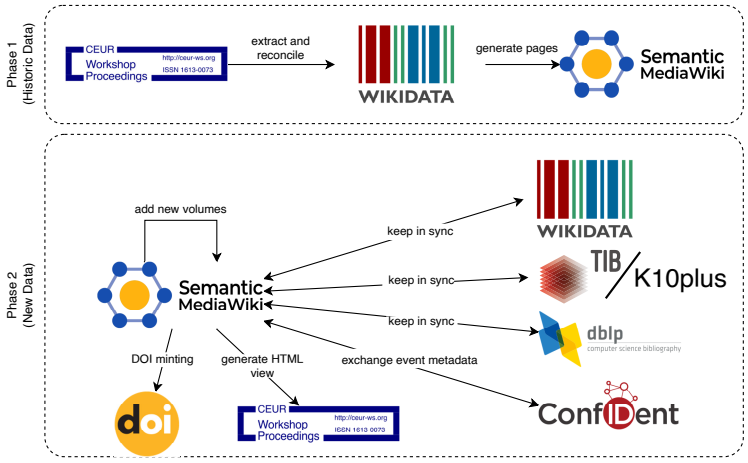

Fig. 9. CEUR-WS Semantification

the "Semantification" project hat has been officially started in February 2023 by the CEUR-WS Editors' team.

Mass creation, editing and disambiguation of relevant scholar entries in Wikidata has to be performed to make these entries available for lookup and reference by the SemPubFlow tool. We intend to build on the Scholia experience with this type of task.

The LLM approach already proven successful for volume homepages needs to also be applied for the much larger data sets of author homepages. An interesting research question is how well the pre processing of PDF files with CERMINE and GROBID is helpful in getting extraction and disambiguation results for paper metadata and how that might influence the efficiency, cost and accuracy.

Given the promising aspects of SemPubFlow, community feedback is essential for its iterative development to better meet the evolving needs of the academic ecosystem. Allowing the integration with other publishing outlets besides CEUR-WS will be a key success factor. Future enhancements will have to focus on increasing the automation and accuracy of metadata extraction and management, while ensuring the tools developed are accessible, user-friendly and well integrated with the publishing workflow.

In promoting metadata-first publishing, a public infrastructure like Wikidata could be used to dynamically generate artifacts, including homepages, CfP webpages, emails, and marketing materials, thus eliminating tedious manual tasks and forming a generic scholary content management system. This approach requires addressing the balance between standardization and individualization, ensuring that while a common framework is maintained for FAIRness and efficiency, the unique aspects of individual publications are not lost. Key to this is the separation of concerns: maintaining distinct layers for metadata storage, con-

tent generation, and presentation, allowing each to evolve independently while cohesively contributing to the scholarly publishing ecosystem.

6.2 Supplemental Material Statement

Source code for the prototypes and semantification tools is available via the repositories of the github ceurws organization[52].

6.3 Acknowledgements

We would like to thank Jakob Voß for helping with the k10plus matching and creating the Wikidata property colocated with (P11633)[53] in due time.

Thomas Hoeren and Jonas Kuiter of ITM, Münster kindly gave hints for legal aspects as summarized in section 4.4.

This paper is dedicated to the memory of CEUR-WS board member Ralf Klamma † January 2023.

This research has been partly funded by a grant of the Deutsche Forschungsgemeinschaft (DFG). [54]

6.4 Declaration of use of Generative AI

References

1. AKhatun: Wikidata Scholarly Articles Subgraph Analysis, `https://wikitech.wikimedia.org/wiki/User:AKhatun/Wikidata_Scholarly_Articles_Subgraph_Analysis`, online; accessed 2023-12-30
2. Bardini, T.: Bootstrapping: Douglas Engelbart, Coevolution, and the Origins of Personal Computing. Stanford University Press, USA (2001)
3. Bast, H., Buchhold, B.: QLever. In: Proceedings of the 2017 ACM Conference on Information and Knowledge Management. pp. 647–656. ACM (Nov 2017). https://doi.org/10.1145/3132847.3132921, `https://doi.org/10.1145/3132847.3132921`
4. Berners-Lee, T., Hendler, J., Lassila, O.: The semantic web. a new form of web content that is meaningful to computers will unleash a revolution of new possibilities. Scientific American **284 (5)**, 34–43 (2001), `https://www.scientificamerican.com/article/the-semantic-web/`
5. Bryl, V., Birukou, A., Eckert, K., Kessler, M.: What's in the proceedings? combining publisher's and researcher's perspectives. In: García Castro, A., Lange, C., Lord, P., Stevens, R. (eds.) 4th Workshop on Semantic Publishing (SePublica). No. 1155 in CEUR Workshop Proceedings, Aachen (2014), `http://ceur-ws.org/Vol-1155#paper-01`
6. CEUR-WS Editor Team: How to submit to ceur workshop proceedings. CEUR Workshop Proceedings (2019), `https://ceur-ws.org/HOWTOSUBMIT.html`, online; accessed 2023-12-28

[52] https://github.com/ceurws
[53] https://www.wikidata.org/wiki/Property:P11633
[54] ConfIDent project; see https://gepris.dfg.de/gepris/projekt/426477583

7. Christina Engelbart: About Bootstrapping . https://www.dougengelbart.org/content/view/226/269/ (2007), online; accessed 12 March 2023
8. Cochinwala, M., Kurien, V., Lalk, G., Shasha, D.: Efficient data reconciliation. Information Sciences **137**(1), 1–15 (2001). https://doi.org/https://doi.org/10.1016/S0020-0255(00)00070-0, https://www.sciencedirect.com/science/article/pii/S0020025500000700
9. Costers, L.: The pica catalogue system – paper 26. In: Proceedings of the IATUL Conferences 1979. pp. 73–77. Purdue University (1979), https://docs.lib.purdue.edu/iatul/1979/papers/26/
10. Cousijn, H., Braukmann, R., Fenner, M., Ferguson, C., van Horik, R., Lammey, R., Meadows, A., Lambert, S.: Connected Research: The Potential of the PID Graph. Patterns (New York, N.Y.) **2**(1), 100180 (2021). https://doi.org/https://doi.org/10.1016/j.patter.2020.100180
11. Cunningham, H., Maynard, D., Bontcheva, K., Tablan, V.: Gate: An architecture for development of robust hlt applications. In: Proceedings of the 40th Annual Meeting on Association for Computational Linguistics. p. 168–175. ACL '02, Association for Computational Linguistics, USA (2002). https://doi.org/10.3115/1073083.1073112, https://doi.org/10.3115/1073083.1073112
12. Cunningham, H., Tablan, V., Roberts, A., Bontcheva, K.: Getting more out of biomedical documents with GATE's full lifecycle open source text analytics. PLoS Computational Biology **9**(2), e1002854 (Feb 2013). https://doi.org/10.1371/journal.pcbi.1002854, https://doi.org/10.1371/journal.pcbi.1002854
13. Fahl, W.: The history of scientific publishing (2023), https://cr.bitplan.com/index.php/The_History_of_Scientific_Publishing, online; accessed 2023-03-15
14. Fahl, W., Eckert, K., Lange, C.: Extracting event metadata from proceedings titles (2022). https://doi.org/10.5281/ZENODO.6568728, https://zenodo.org/record/6568728
15. Franken, J., Birukou, A., Eckert, K., Fahl, W., Hauschke, C., Lange, C.: Persistent identification for conferences. Data Science Journal **21** (2022). https://doi.org/10.5334/dsj-2022-011
16. Ganseman, J.: Refactoring a library's legacy catalog: a case study. In: IAML Congress 2015. IAML (2015), http://wiki.muziekcollecties.be/images/IAML2015_JG.pdf
17. GO FAIR International Support and Coordination Office: Fair principles (2019), https://www.go-fair.org/fair-principles/
18. Hogan, A., Blomqvist, E., Cochez, M., d'Amato, C., de Melo, G., Gutierrez, C., Kirrane, S., Gayo, J.E.L., Navigli, R., Neumaier, S., Ngonga Ngomo, A.C., Polleres, A., Rashid, S.M., Rula, A., Schmelzeisen, L., Sequeda, J., Staab, S., Zimmermann, A.: Knowledge graphs. ACM Computing Surveys **54**(4), 1–37 (jul 2021). https://doi.org/10.1145/3447772, https://doi.org/10.1145/F3447772
19. Käfer, T., Abdelrahman, A., Umbrich, J., O'Byrne, P., Hogan, A.: Observing linked data dynamics. In: The Semantic Web: Semantics and Big Data. pp. 213–227. Springer Berlin Heidelberg, Berlin, Heidelberg (2013). https://doi.org/10.1007/978-3-642-38288-8_15
20. Kim, J.: Evaluating author name disambiguation for digital libraries: a case of DBLP. Scientometrics **116**(3), 1867–1886 (Jun 2018). https://doi.org/10.1007/s11192-018-2824-5, https://doi.org/10.1007/s11192-018-2824-5

21. Kolchin, M., Cherny, E., Kozlov, F., Shipilo, A., Kovriguina, L.: CEUR-WS-LOD: Conversion of CEUR-WS workshops to linked data. In: Semantic Web Evaluation Challenges, pp. 142–152. Springer International Publishing (2015). https://doi.org/10.1007/978-3-319-25518-7_12, https://doi.org/10.1007/978-3-319-25518-7_12
22. Kolchin, M., Kozlov, F.: A template-based information extraction from web sites with unstable markup. In: Semantic Web Evaluation Challenge, pp. 89–94. Communications in Computer and Information Science, Springer International Publishing (2014). https://doi.org/10.1007/978-3-319-12024-9_11, https://doi.org/10.1007/978-3-319-12024-9_11
23. Krötzsch, M., Vrandečić, D.: Semantic MediaWiki. In: Foundations for the Web of Information and Services, pp. 311–326. Springer Berlin Heidelberg (2011). https://doi.org/10.1007/978-3-642-19797-0_16, https://doi.org/10.1007/2F978-3-642-19797-0_16
24. Kunze, J., Bermès, E.: The ark identifier scheme (2022), https://www.ietf.org/archive/id/draft-kunze-ark-36.html
25. Lange, C., Di Iorio, A.: Semantic publishing challenge – assessing the quality of scientific output. In: Communications in Computer and Information Science, pp. 61–76. Springer International Publishing (2014). https://doi.org/10.1007/978-3-319-12024-9_8, https://doi.org/10.1007/978-3-319-12024-9_8
26. Ley, M.: DBLP. Proceedings of the VLDB Endowment **2**(2), 1493–1500 (Aug 2009). https://doi.org/10.14778/1687553.1687577, https://doi.org/10.14778/1687553.1687577
27. Lopez, P.: GROBID: Combining automatic bibliographic data recognition and term extraction for scholarship publications. In: Research and Advanced Technology for Digital Libraries, pp. 473–474. Springer Berlin Heidelberg (2009). https://doi.org/10.1007/978-3-642-04346-8_62, https://doi.org/10.1007/978-3-642-04346-8_62
28. Manske, M.: QuickStatements (2016), https://www.wikidata.org/wiki/Help:QuickStatements
29. Milicka, M., Burget, R.: Information extraction from web sources based on multi-aspect content analysis. In: Semantic Web Evaluation Challenges, pp. 81–92. Springer International Publishing (2015). https://doi.org/10.1007/978-3-319-25518-7_7, https://doi.org/10.1007/F978-3-319-25518-7_7
30. Nielsen, F.Å., Mietchen, D., Willighagen, E.: Scholia, scientometrics and wikidata. In: Blomqvist, E., Hose, K., Paulheim, H., Ławrynowicz, A., Ciravegna, F., Hartig, O. (eds.) The Semantic Web: ESWC 2017 Satellite Events. pp. 237–259. Springer International Publishing, Cham (2017). https://doi.org/10.1007/978-3-319-70407-4_36
31. OECD: OECD Glossary of Statistical Terms. OECD (sep 2008). https://doi.org/10.1787/9789264055087-en, https://doi.org/10.1787/9789264055087-en
32. Pohl, K., Rupp, C.: Requirements engineering fundamentals. Rocky Nook, Santa Barbara, CA, 2 edn. (May 2015)
33. Priem, J., Piwowar, H., Orr, R.: OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts. 26th International Conference on Science, Technology and Innovation Indicators (STI 2022) (2022). https://doi.org/10.5281/ZENODO.6936227, https://zenodo.org/record/6936227

34. Sateli, B., Witte, R.: Automatic construction of a semantic knowledge base from CEUR workshop proceedings. In: Semantic Web Evaluation Challenges, pp. 129–141. Springer International Publishing (2015). https://doi.org/10.1007/978-3-319-25518-7_11, https://doi.org/10.1007/978-3-319-25518-7_11
35. Sateli, B., Witte, R.: From papers to triples: An open source workflow for semantic publishing experiments. In: Semantics, Analytics, Visualization. Enhancing Scholarly Data, pp. 39–44. Springer International Publishing (2016). https://doi.org/10.1007/978-3-319-53637-8_5, https://doi.org/10.1007/978-3-319-53637-8_5
36. Sollins, K.R.: Pervasive persistent identification for information centric networking. In: Proceedings of the second edition of the ICN workshop on Information-centric networking. pp. 1–6. ACM, New York, NY, USA (2012). https://doi.org/10.1145/2342488.2342490
37. Subramanian, S., King, D., Downey, D., Feldman, S.: S2and: A benchmark and evaluation system for author name disambiguation. In: 2021 ACM/IEEE Joint Conference on Digital Libraries (JCDL). pp. 170–179. IEEE (Sep 2021). https://doi.org/10.1109/jcdl52503.2021.00029, https://doi.org/10.1109/jcdl52503.2021.00029
38. Tkaczyk, D., Szostek, P., Fedoryszak, M., Dendek, P.J., Bolikowski, Ł.: CERMINE: automatic extraction of structured metadata from scientific literature. International Journal on Document Analysis and Recognition (IJDAR) **18**(4), 317–335 (Jul 2015). https://doi.org/10.1007/s10032-015-0249-8, https://doi.org/10.1007/s10032-015-0249-8
39. Vrandečić, D., Krötzsch, M.: Wikidata. Communications of the ACM **57**(10), 78–85 (Sep 2014). https://doi.org/10.1145/2629489, https://doi.org/10.1145/2629489
40. Wang, K., Shen, Z., Huang, C., Wu, C.H., Dong, Y., Kanakia, A.: Microsoft Academic Graph: When experts are not enough. Quantitative Science Studies **1**(1), 396–413 (Feb 2020). https://doi.org/10.1162/qss_a_00021, https://doi.org/10.1162/qss_a_00021
41. Wiermann, B.: K10plus – Zehn Bundesländer in einem Bibliothekssystem (2019), https://blog.slub-dresden.de/beitrag/2019/03/27/k10plus-zehn-bundeslaender-in-einem-bibliothekssystem
42. Wilkinson, M.D., Dumontier, M., Aalbersberg, I.J., Appleton, G., Axton, M., Baak, A., Blomberg, N., Boiten, J.W., da Silva Santos, L.B., Bourne, P.E., Bouwman, J., Brookes, A.J., Clark, T., Crosas, M., Dillo, I., Dumon, O., Edmunds, S., Evelo, C.T., Finkers, R., Gonzalez-Beltran, A., Gray, A.J., Groth, P., Goble, C., Grethe, J.S., Heringa, J., 't Hoen, P.A., Hooft, R., Kuhn, T., Kok, R., Kok, J., Lusher, S.J., Martone, M.E., Mons, A., Packer, A.L., Persson, B., Rocca-Serra, P., Roos, M., van Schaik, R., Sansone, S.A., Schultes, E., Sengstag, T., Slater, T., Strawn, G., Swertz, M.A., Thompson, M., van der Lei, J., van Mulligen, E., Velterop, J., Waagmeester, A., Wittenburg, P., Wolstencroft, K., Zhao, J., Mons, B.: The FAIR guiding principles for scientific data management and stewardship. Scientific Data **3**(1) (mar 2016). https://doi.org/10.1038/sdata.2016.18, https://doi.org/10.1038/data.2016.18
43. Wolfgang Fahl: Smwcon fall 2020/backend infrastructure experience exchange e.g. sqlstore, elasticsearch, blazegraph, virtuoso, jena (2020), https://www.semantic-mediawiki.org/wiki/SMWCon_Fall_2020/Backend_infrastructure_experience_exchange_e.g._SqlStore,_Elasticsearch,_Blazegraph,_Virtuoso,_Jena, online; accessed 2023-12-30

44. Yu, C., Zhang, C., Wang, J.: Extracting body text from academic PDF documents for text mining. CoRR **abs/2010.12647** (2020), https://arxiv.org/abs/2010.12647